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Bayesian optimization of sparsity ratios in model compression

Abstract

One embodiment of a method includes determining, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model. The method further includes selecting, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the machine learning model within a search space bounded by the first sparsity ratio. The method further includes generating a compressed version of the machine learning model having the second sparsity ratio.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of the U.S. Provisional application titled "RESOURCE-EFFICIENT BAYESIAN MODEL COMPRESSION," filed on Aug. 26, 2019, and having Ser. No. 62/891,897. The subject matter of this application is hereby incorporated herein by reference in its entirety.

BACKGROUND

Field of the Various Embodiments

(1) Embodiments of the present disclosure relate generally to compression of machine learning models, and more specifically, to Bayesian optimization of sparsity ratios in model compression. Description of the Related Art

(2) Deep neural networks commonly have a large number of computation-heavy layers such as convolutional layers and/or fully-connected layers. Such neural networks are also frequently trained and deployed using full-precision arithmetic. As a result, executing DNNs can involve significant complexity, latency, and memory consumption.

(3) A number of techniques can be employed to reduce the memory footprint and inference time of deep neural networks. For example, parameters and/or activations of a neural network can be quantized to reduce the bit width of the parameters (e.g., from 32-bit floating point to 8-bit floating point) and/or convert certain matrix multiplication operations for calculating the activations into logical operations. Further, a weight in the neural network can be pruned (i.e., removed) when the weight has an absolute value and/or an effect on the training error of the neural network that falls below a threshold.

(4) Existing techniques for compressing neural networks require the explicit specification of a target sparsity ratio representing the proportion of parameters in a pruned model with zero values compared with the original model. Alternatively, the existing techniques perform manual exploration of the search space for pruning or compressing a neural network. This manual exploration and/or selection of the amount of compression is computationally intensive and inconvenient, as the optimal sparsity ratio depends on the neural network, target platform, and optimization objective, and each trial in the exploration requires retraining the compressed model to convergence.

(5) As the foregoing illustrates, what is needed in the art are more efficient techniques for determining optimal sparsity ratios for compressing machine learning models.

SUMMARY

(6) One embodiment of the present invention sets forth a technique for compressing a machine

learning model. The technique includes determining, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model. The technique further includes selecting, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the machine learning model within a search space bounded by the first sparsity ratio. The technique further includes generating a compressed version of the machine learning model having the second sparsity ratio.

(7) One technological advantage of the disclosed techniques is the ability to infer a sparsity ratio that meets constraints and/or objectives associated with compressing the machine learning model instead of requiring a user to manually specify a sparsity ratio for compressing the machine learning model. Consequently, the disclosed techniques provide technological improvements in computer systems, applications, and/or techniques for compressing and/or executing machine learning models.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the various embodiments can be understood in detail, a more particular description of the inventive concepts, briefly summarized above, may be had by reference to various embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of the inventive concepts and are therefore not to be considered limiting of scope in any way, and that there are other equally effective embodiments.

(2) FIG. 1 illustrates a system configured to implement one or more aspects of various embodiments.

(3) FIG. 2 is a more detailed illustration of the Bayesian optimizer and compression engine of FIG. 1, according to one or more aspects of various embodiments.

(4) FIG. 3 is an example set of plots of iterations performed by the Bayesian optimizer of FIG. 2, according to one or more aspects of various embodiments.

(5) FIG. 4 is a flow diagram of method steps for selecting a sparsity ratio for compressing a machine learning model, according to one or more aspects of various embodiments.

(6) FIG. 5 is a flow diagram of method steps for compressing a machine learning model, according to one or more aspects of various embodiments.

DETAILED DESCRIPTION

(7) In the following description, numerous specific details are set forth to provide a more thorough understanding of the various embodiments. However, it will be apparent to one of skilled in the art that the inventive concepts may be practiced without one or more of these specific details.

(8) System Overview

(9) FIG. 1 illustrates a computing device **100** configured to implement one or more aspects of various embodiments. In one embodiment, computing device **100** may be a desktop computer, a laptop computer, a smart phone, a personal digital assistant (PDA), tablet computer, or any other type of computing device configured to receive input, process data, and optionally display images, and is suitable for practicing one or more embodiments. Computing device **100** is configured to run a Bayesian optimizer **122** and compression engine **124** that reside in a memory **116**. It is noted that the computing device described herein is illustrative and that any other technically feasible configurations fall within the scope of the present disclosure. For example, multiple instances of Bayesian optimizer **122** and compression engine **124** may execute on a set of nodes in a distributed system to implement the functionality of computing device **100**.

(10) In one embodiment, computing device **100** includes, without limitation, an interconnect (bus) **112** that connects one or more processing units **102**, an input/output (I/O) device interface **104**

coupled to one or more input/output (I/O) devices **108**, memory **116**, a storage **114**, and a network interface **106**. Processing unit(s) **102** may be any suitable processor implemented as a central processing unit (CPU), a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA), an artificial intelligence (AI) accelerator, any other type of processing unit, or a combination of different processing units, such as a CPU configured to operate in conjunction with a GPU. In general, processing unit(s) **102** may be any technically feasible hardware unit capable of processing data and/or executing software applications. Further, in the context of this disclosure, the computing elements shown in computing device **100** may correspond to a physical computing system (e.g., a system in a data center) or may be a virtual computing instance executing within a computing cloud.

(11) In one embodiment, I/O devices **108** include devices capable of providing input, such as a keyboard, a mouse, a touch-sensitive screen, and so forth, as well as devices capable of providing output, such as a display device. Additionally, I/O devices **108** may include devices capable of both receiving input and providing output, such as a touchscreen, a universal serial bus (USB) port, and so forth. I/O devices **108** may be configured to receive various types of input from an end-user (e.g., a designer) of computing device **100**, and to also provide various types of output to the end-user of computing device **100**, such as displayed digital images or digital videos or text. In some embodiments, one or more of I/O devices **108** are configured to couple computing device **100** to a network **110**.

(12) In one embodiment, network **110** is any technically feasible type of communications network that allows data to be exchanged between computing device **100** and external entities or devices, such as a web server or another networked computing device. For example, network **110** may include a wide area network (WAN), a local area network (LAN), a wireless (WiFi) network, and/or the Internet, among others.

(13) In one embodiment, storage **114** includes non-volatile storage for applications and data, and may include fixed or removable disk drives, flash memory devices, and CD-ROM, DVD-ROM, Blu-Ray, HD-DVD, or other magnetic, optical, or solid state storage devices. Bayesian optimizer **122** and compression engine **124** may be stored in storage **114** and loaded into memory **116** when executed.

(14) In one embodiment, memory **116** includes a random access memory (RAM) module, a flash memory unit, or any other type of memory unit or combination thereof. Processing unit(s) **102**, I/O device interface **104**, and network interface **106** are configured to read data from and write data to memory **116**. Memory **116** includes various software programs that can be executed by processor(s) **102** and application data associated with said software programs, including Bayesian optimizer **122** and compression engine **124**.

(15) In some embodiments, Bayesian optimizer **122** and compression engine **124** include functionality to improve the runtime performance of a machine learning model by compressing the machine learning model. For example, the machine learning model includes a deep neural network (DNN) with a large number of neurons, parameters, and/or layers. As a result, inference by the DNN requires significant consumption of computational and memory resources. To improve the inference time, memory footprint, and/or other runtime attributes of the machine learning model, Bayesian optimizer **122** and/or compression engine **124** create a compressed version of the machine learning model by selectively pruning a subset of weights in the DNN and/or quantizing the weights. In turn, the compressed version incurs significantly less resource overhead and latency than the uncompressed model without appreciable loss of accuracy.

(16) More specifically, Bayesian optimizer **122** and compression engine **124** include functionality to optimize the compression of a machine learning model according to predefined and/or tunable objectives and/or constraints. Continuing with the above example, Bayesian optimizer **122** receives, from a user, an objective to be optimized in compressing the machine learning model, a lower limit on the loss of accuracy associated with compressing the machine learning model,

and/or a compression scheme. Bayesian optimizer **122** and compression engine **124** then iteratively sample different sparsity ratios for compressing the machine learning model using the compression scheme. Finally, Bayesian optimizer **122** and compression engine **124** produce a compressed version of the machine learning model having a sparsity ratio that best optimizes the objective while meeting the constraint represented by the limit on the accuracy loss. Various embodiments of Bayesian optimizer **122** and compression engine **124** are described in further detail below with respect to FIG. 2.

(17) Bayesian Optimization of Sparsity Ratios in Model Compression

(18) FIG. 2 is a more detailed illustration of Bayesian optimizer **122** and compression engine **124** of FIG. 1, according to various embodiments. In the embodiment shown, Bayesian optimizer **122** samples sparsity ratios **218-220** for generating a compressed machine learning model **226** from a reference machine learning model **222** (e.g., an uncompressed machine learning model and/or a compressed machine learning model that is to be compressed further) based on a predefined accuracy loss limit **206**, objective function **208**, and compression scheme **240**. For example, Bayesian optimizer **122** receives accuracy loss limit **206**, objective function **208**, and/or compression scheme **240** from a user via function calls, command line parameters, a graphical user interface (GUI), and/or another type of interface or communication mechanism.

(19) In some embodiments, a sparsity ratio (e.g., sparsity ratios **218-220**) represents the proportion of parameters that have been pruned (e.g., replaced with zero values) in compressed machine learning model **226** compared with the original reference machine learning model **222**. The sparsity ratio also, or instead, reflects a ratio of the size of compressed machine learning model **226** to the size of the original reference machine learning model **222**. For example, the sparsity ratio may be obtained by subtracting the density ratio of compressed machine learning model **226** from 1, where the density ratio equals the size of compressed machine learning model **226** divided by the size of reference machine learning model **222**. In general, the sparsity ratio indicates the reduction in size of compressed machine learning model **226** when compared with the original, reference machine learning model **222**.

(20) Accuracy loss limit **206** represents a lower bound on the reduction in accuracy caused by generating compressed machine learning model **226** from reference machine learning model **222**. For example, accuracy loss limit **206** is expressed by the user as a percentage or proportion of the accuracy of reference machine learning model **222**.

(21) In some embodiments, objective function **208** includes a black-box function to be maximized or minimized. For example, a user specifies objective function **208** as an inference throughput (e.g., a rate at which compressed machine learning model **226** performs evaluation), memory footprint, latency (e.g., the amount of time required by compressed machine learning model **226** to perform evaluation), a domain-specific objective (e.g., “smoothness” or frame rate of video output), and/or another value that can be measured or calculated from the execution or output of compressed machine learning model **226**. The user additionally specifies minimization or maximization of objective function **208** as an objective associated with compressed machine learning model **226**.

(22) Compression scheme **240** includes a type of compression to be applied to produce compressed machine learning model **226**. For example, a user selects compression scheme **240** from a list that includes, but is not limited to, quantization of parameters (e.g., from 32-bit floating point to 16-bit floating point), unstructured magnitude-based pruning of weights in a DNN, pruning of neurons in the DNN, pruning of blocks of various dimensions in the DNN, and/or low-rank tensor factorization. In another example, the user programmatically expresses compression scheme **240** as the composition of multiple compression schemes with control flow, iteration, recursion, and/or other imperative statements. These statements allow for selective compression of layers and/or other portions of reference machine learning model **222** based on properties of reference machine learning model **222** and/or the target hardware on which compressed machine learning model **226** is to be deployed.

(23) In one or more embodiments, Bayesian optimizer 122 and compression engine 124 perform a number of iterations 202-204 that sample different sparsity ratios 218 for producing compressed machine learning model 226 from reference machine learning model 222 using compression scheme 240. During each iteration, Bayesian optimizer 122 selects a sparsity ratio for compressed machine learning model 226, and compression engine 124 uses compression scheme 240 to produce compressed machine learning model 226 from reference machine learning model 222 in a way that adheres to the selected sparsity ratio. Compression engine 124 also measures an accuracy 236, performance 238, and/or another value related to the execution or output of compressed machine learning model 226 and provides the measurement to Bayesian optimizer 122. Bayesian optimizer uses measurements from compression engine 124 to update a probabilistic belief about the output as a function of sparsity ratio in compressed machine learning model 226 and uses an acquisition function to select the next sparsity ratio to sample in the next iteration. At the end of iterations 202-204, Bayesian optimizer 122 and compression engine 124 produce compressed machine learning model 226 having a sparsity ratio that optimizes objective function 208 while conforming to the constraint represented by accuracy loss limit 206.

(24) During each iteration performed by Bayesian optimizer 122, compression engine 124 generates compressed machine learning model 226 having the sparsity ratio selected by Bayesian optimizer 122 for that iteration. As shown, such compression involves a learning step 228 that trains reference machine learning model 222 with a quadratic regularization term 232 and a compression step 230 that converts the trained reference machine learning model 222 from learning step 228 into compressed machine learning model 226 version with compressed parameters 234.

(25) For example, compression of reference machine learning model 222 by compression engine 124 is formulated as a constrained optimization problem with the following representation:

$$\min.\text{sub.}w,\theta L(w)s.t.w=D(\theta)$$

In the above representation, $L(\cdot)$ denotes a loss function used to train reference machine learning model 222, and $w \in \mathbb{R}^{\text{sup.}P}$ represents uncompressed parameters of reference machine learning model 222 (e.g., neural network weights). A decompression mapping

$D:\theta \in \mathbb{R}^{\text{sup.}P} \rightarrow w \in \mathbb{R}^{\text{sup.}n}$ maps compressed parameters 234 θ of compressed machine learning model 226 to the uncompressed parameters.

(26) The above formulation supports a number of compression schemes. For example, pruning is defined as $w=D(\theta)=\theta$, where w is real and θ is constrained to have fewer nonzero values by removing (zeroing out) lower magnitude weights. In another example, low-precision approximation (i.e., quantization) defines a per-parameter constraint $w_{\text{sub.}i}=\theta_{\text{sub.}i}$, where $w_{\text{sub.}i}$ is a higher-precision representation of a weight in reference machine learning model 222 and $\theta_{\text{sub.}i}$ is a lower-precision representation of the same weight in compressed machine learning model 226.

(27) A compression mapping behaves similarly to the inverse of D :

$$\text{Math.}_C(w; \cdot) = \arg\min \text{Math. } w - \text{Math.}^2 s.t. C(\cdot) \leq$$

In the above expression, $C:\mathbb{R}^{\text{sup.}n} \rightarrow \mathbb{R}^{\text{sup.}+}$ represents a cost function that promotes sparsity in compressed parameters 234, where $C(0)=0$ and $C(w)>0$ if $w \neq 0$. κ represents a constraint on the number or complexity of compressed parameters 234, and the “ $\leq$ ” superindex refers to the constraint form of the compression mapping.

(29) In turn, the constrained optimization problem can be expressed using the following alternative representation:

$$\min.\text{sub.}w,\theta L(w)s.t. C(\theta) \leq \kappa, w=\theta$$

The constrained optimization problem can then be optimized via quadratic regularization term 232 and/or an augmented-Lagrangian (AL) method:

$$(30) Q(w, \cdot, \cdot) = L(w) + \frac{1}{2} \text{Math. } w - \text{Math.}^2 s.t. C(\cdot) \leq$$

$$L_A(w, \cdot, \cdot; \cdot) = L(w) + \frac{1}{2} \text{Math. } w - \text{Math.}^2 - T(w - \cdot) s.t. C(\cdot) \leq$$

The first equation uses a value of

$$(31) \quad \frac{1}{2} \|\mathbf{w} - \mathbf{0}\|_2^2$$

for quadratic regularization term **232** to optimize Q over $(\mathbf{w}, \theta)$ while driving $\mu \rightarrow \infty$, so that the equality constraints are satisfied in the limit. The second equation uses the AL method to alternate optimizing $L(\mathbf{w}, \theta)$ over $(\mathbf{w}, \theta)$ with updating $\lambda \leftarrow -\lambda - \mu(\mathbf{w} - \theta)$ while driving $\mu \rightarrow \infty$.

(32) To optimize the above equations over the variables $(\mathbf{w}, \theta)$, compression engine **124** applies alternating optimization using learning step **228** and compression step **230**. During learning step **228**, compression engine **124** performs the following optimization using the following equation:

$$(33) \quad \min_{\mathbf{w}} L(\mathbf{w}) + \frac{1}{2} \|\mathbf{w} - \mathbf{0}\|_2^2$$

The second term in the above equation represents quadratic regularization term **232**, which is used in training of reference machine learning model **226** to pull some parameters to zero and other parameters to a nonzero value.

(34) If the AL method is used in the optimization, compression engine **124** performs learning step **228** using the following equation:

$$(35) \quad \min_{\mathbf{w}} L(\mathbf{w}) + \frac{1}{2} \|\mathbf{w} - \mathbf{0}\|_2^2 + \frac{1}{2} \|\mathbf{w} - \theta\|_2^2$$

(36) During compression step **230**, the compression mapping above can be solved for various costs C , such as (but not limited to) $l_{sub,0}$, $l_{sub,1}$, and $l_{sub,2,sup,2}$. Each of these costs can be solved by projecting on $l_{sub,p}$ balls. When $\mathbf{w}$ is in the ball, $\theta = \mathbf{w}$. Otherwise, $l_{sub,0}$ leaves the top- κ weights unchanged and prunes the rest, $l_{sub,1}$ shrinks on average the top- κ weights (where κ depends on $\mathbf{w}$ and κ) and prunes the rest, and $l_{sub,2,sup,2}$ shrinks all weights by normalizing $\mathbf{w}$.

(37) Moreover, compression step **230** “marks” parameters to be compressed by setting the $\theta_{sub,1}$ values. A subsequent learning step **228** then updates the $\mathbf{w}_{sub,i}$ values according to the loss function and the marked $\theta_{sub,1}$ values. As compression engine **124** iteratively alternates between learning step **228** and compression step **230**, different sets of marked weights are explored until a set of marked weights is found for which $\mathbf{w}_{sub,i} \rightarrow \mathbf{0}$ as $\mu \rightarrow \infty$, which results in pruning of those weights. Consequently, compression engine **124** produces compressed machine learning model **226** with a given sparsity ratio and a locally optimal loss of accuracy from reference machine learning model **222**.

(38) In some embodiments, Bayesian optimizer **122** performs a first set of iterations **202** that sample sparsity ratios **218** for producing compressed machine learning model **226** to identify a highest sparsity ratio that falls within accuracy loss limit **206**. Bayesian optimizer **122** then sets the identified sparsity ratio as a boundary of a search space **224** for a second set of iterations **204**.

During the second set of iterations **202**, Bayesian optimizer **122** samples additional sparsity ratios **220** for producing compressed machine learning model **226** within search space **244** to identify a sparsity ratio that produces the best performance **238** in optimizing objective function **208**.

(39) In one embodiment, Bayesian optimizer **122** utilizes a Gaussian Process (GP) prior distribution for a given black box function $f(\mathbf{x})$, which can include accuracy **236** and/or objective function **208**. The GP includes a distribution over functions that is specified by a mean function $m: \mathbf{X} \rightarrow \mathbb{R}$ and a covariance function $K: \mathbf{X} \times \mathbf{X} \rightarrow \mathbb{R}$. As Bayesian optimizer **122** and compression engine **124** accumulate observations $D_{sub,1:t} = \{\mathbf{x}_{sub,1:t}, y_{sub,1:t}\}$, Bayesian optimizer **122** combines the prior distribution $P(f)$ with the likelihood function $P(D_{sub,1:t}|f)$ to produce the posterior distribution $P(f|D_{sub,1:t}) \propto P(D_{sub,1:t}|f)P(f)$.

(40) During each iteration in iterations **202**, compression engine **124** produces compressed machine learning model **226** at a sparsity ratio (e.g., sparsity ratios **218**) sampled by Bayesian optimizer **122** for that iteration. Compression engine **124** also determines accuracy **236** of compressed machine learning model **226**, and Bayesian optimizer **122** updates the GP prior with the calculated accuracy

236 before optimizing an acquisition function over the GP prior to determine the next sparsity ratio to sample in a successive iteration.

(41) At the end of iterations **202**, Bayesian optimizer **122** sets the boundaries of search space **224** for iterations **204** to range from a sparsity ratio of 0 to the sparsity ratio identified by iterations **202** as having substantially the same accuracy **236** as accuracy loss limit **206**. During each iteration in iterations **204**, compression engine **124** produces compressed machine learning model **226** at a sparsity ratio (e.g., sparsity ratios **220**) sampled by Bayesian optimizer **122** for that iteration. Compression engine **124** also determines performance **238** of compressed machine learning model **226** as the value of objective function **208** for compressed machine learning model **226**, and Bayesian optimizer **122** updates the GP prior with performance **238** before optimizing an acquisition function over the GP prior to determine the next sparsity ratio to sample in a subsequent iteration.

(42) Bayesian optimizer also uses sampling criteria represented by one or more acquisition functions **214-216** $a: X \rightarrow \mathbb{R}$ to guide the selection of the next sparsity ratio to sample. These acquisition functions **214-216** depend on the previous observations $\{x_{\text{sub},n}, y_{\text{sub},n}\}$ and the GP hyperparameters ρ . Under the GP prior, each acquisition function **214** or **216** depends on a predictive mean function $\mu(x; \{x_{\text{sub},n}, y_{\text{sub},n}\}, \rho)$ and predictive variance function $\sigma^2(x; \{x_{\text{sub},n}, y_{\text{sub},n}\}, \Sigma)$.

(43) Bayesian optimizer **122** additionally selects sparsity ratios **218-220** using a combination of random sampling **210-212** and sampling criteria based on acquisition functions **214-216** for each set of iterations **202-204**. For example, Bayesian optimizer **122** starts iterations **202** with one or more rounds of random sampling **210** of sparsity ratios **218**. Bayesian optimizer **122** obtains values of accuracy **236** returned by compression engine **124** for the randomly sampled sparsity ratios **218** and updates the GP prior for accuracy **236** using the values. Bayesian optimizer **122** then performs additional iterations **202** that sample sparsity ratios **218** according to acquisition function **214** associated with the GP prior for accuracy **236**.

(44) Continuing with the above example, Bayesian optimizer **122** similarly starts iterations **204** with one or more rounds of random sampling **212** of sparsity ratios **220**. Bayesian optimizer **122** obtains values of performance **238** returned by compression engine **124** for the randomly sampled sparsity ratios **220** and updates the GP prior for objective function **208** using the values. Bayesian optimizer **122** then performs additional iterations **204** that sample sparsity ratios **220** according to acquisition function **216** associated with the GP prior for objective function **208**.

(45) The operation of Bayesian optimizer **122** is illustrated using the following example pseudocode:

(46) TABLE-US-00001 Input: w, ϵ Output: s^* AcqFn $\leftarrow$ ILS-UCB($L = w_{\text{sub},\text{acc}} - \epsilon, s = (0, 1)$)
 $S_{\text{sub},\text{acc}} \leftarrow$ BayesOpt($B_{\text{sub},f} = L - C, \text{AcqFn}$) AcqFn $\leftarrow$ GP-UCB($s = (0, s_{\text{sub},\text{acc}})$) $s^* \leftarrow$
BayesOpt($B_{\text{sub},f} = f, \text{AcqFn}$) BayesOpt Input: $B_{\text{sub},f}, \text{AcqFn}$ Output: s GP $\leftarrow$ GP-
Regressor.initialize() for $t \leftarrow 0, 1, 2, \dots$ do $s_{\text{sub},t} \leftarrow \text{argmax}_{s \in S} \text{AcqFn}(s \mid D_{\text{sub},1:t-1})$
 $y_{\text{sub},t} \leftarrow f(s_{\text{sub},t})$ $D_{\text{sub},1:t} \leftarrow \{D_{\text{sub},1:t-1}, (s_{\text{sub},t}, y_{\text{sub},t})\}$ GP.Update($D_{\text{sub},1:t}$) if $t >$
0 and $s_{\text{sub},t} == s_{\text{sub},t-1}$ then return $s_{\text{sub},t}$ end if end for

(47) In the above pseudocode, input to Bayesian optimizer **122** includes a reference machine learning model **222** w and accuracy loss limit **206** ϵ . The output of Bayesian optimizer **122** s^* represents a sparsity ratio for compressed machine learning model **226** with the best performance **238** in optimizing objective function **208** and a value of accuracy **236** that falls within accuracy loss limit **206**.

(48) More specifically, the above pseudocode uses an implicit level set upper confidence bound (ILS-UCB) acquisition function **214** for iterations **202**, where the level set L is defined by subtracting ϵ from the accuracy of reference machine learning model **222**. In turn, iterations **202** prioritize sampling of sparsity ratios **218** near the level set to localize the sparsity that meets accuracy loss limit **206**, which reduces the total number of measurements and time required to

achieve an optimal sparsity ratio $s_{\text{sub},\text{acc}}$ for accuracy loss limit **206**.

(49) For example, the ILS-UCB acquisition function **214** used to select sparsity ratios **218** during iterations **202** includes the following representation:

$$(50) \quad x_t = \underset{x \in \mathcal{X}}{\operatorname{argmax}} (1 - \gamma) f(x) - \gamma \mathcal{L}(x) - \frac{\gamma}{2} \sigma^2(x)$$

In the above representation, γ represents a tuning parameter between 0 and 1 that balances the contribution of the predictive mean of acquisition function **214** with the contribution of the predictive variance of the acquisition function to the sample in iteration t .

(51) Next, the pseudocode uses a search space constrained by $(0, s_{\text{sub},\text{acc}})$ and an acquisition function of GP-UCB as acquisition function **216** for iterations **204**. The pseudocode then executes iterations **204** to sample sparsity ratios **220** in a way that maximizes objective function **208**.

Alternatively, an different acquisition function **216** of GP lower confidence bound (GP-LCB) can be used to sample sparsity ratios **220** in a way that minimizes objective function **208**. When acquisition function **216** selects the same sparsity ratio in two consecutive iterations **204**, iterations **204** are discontinued, and the selected sparsity ratio is returned as the optimal sparsity ratio s^* .

(52) FIG. 3 is an example set of plots **302-304** of iterations performed by Bayesian optimizer **122** of FIG. 2, according to one or more aspects of various embodiments. More specifically, plot **302** shows a prediction **306** by Bayesian optimizer **122** of a black box function that varies with the sparsity ratio of compressed machine learning model **226**, and plot **304** shows an acquisition function **322** used by Bayesian optimizer **122** to select a next sample **320** of the black box function to evaluate.

(53) Within plot **302**, prediction **306** represents the posterior predictive mean of a GP prior that is conditioned on previous samples **312-318** of the black box function. Plot **302** also includes a shaded region **310** surrounding prediction **306** that represents the 95% confidence interval of the mean. Plot **302** additionally shows target **308** values of the black box function to be predicted by Bayesian optimizer **122** via samples **312-318**.

(54) In one embodiment, samples **312-318** include four observations of accuracy **236** of compressed machine learning model **226** at various sparsity ratios **218** sampled by Bayesian optimizer **122** over four iterations **202**. For example, samples **312-314** represent two randomly sampled sparsity ratios that are used to “seed” the GP prior, and samples **316-318** represent sparsity ratios that are selected by Bayesian optimizer **122** based on acquisition function **322**. After the black box function is evaluated for samples **316-318**, Bayesian optimizer **122** updates the GP prior to produce an updated prediction **306** that better approximates the black box function. Bayesian optimizer **122** then uses the updated prediction **306** to construct or update an ILS-UCB acquisition function **322** and selects the next sample **320** for a successive iteration as the maximum of acquisition function **322**.

(55) For example, Bayesian optimizer **122** uses ILS-UCB to prioritize sampling in areas near a level set where the mean of the GP prior is closest to a threshold **324** representing accuracy loss limit **206** for compressed machine learning model **226**. After a number of iterations **202**, Bayesian optimizer **122** identifies sample **318** as having a sparsity ratio that produces a compressed machine learning model **226** with accuracy **236** that is equal to accuracy loss limit **206**. In turn, Bayesian optimizer **122** performs additional iterations **204** that sample sparsity ratios **220** within search space **224** that is bounded by the sparsity ratio of sample **318** to select another sparsity ratio for producing compressed machine learning model **226** that optimizes a user-specified objective function **208** while meeting accuracy loss limit **206**.

(56) FIG. 4 is a flow diagram of method steps for compressing a machine learning model, according to one or more aspects of various embodiments. Although the method steps are described in conjunction with the systems of FIGS. 1 and 2, persons skilled in the art will understand that any system configured to perform the method steps in any order falls within the scope of the present disclosure.

(57) As shown, Bayesian optimizer **122** receives **402** a predefined objective function, a limit on an accuracy loss caused by compressing the machine learning model, and/or a compression scheme from a user. For example, the objective function includes an evaluation rate for the machine learning model, memory footprint, evaluation time for the machine learning model, domain-specific objective (e.g., “smoothness” or frame rate of video output), and/or another value that can be measured or calculated from the execution or output of the machine learning model. The user additionally specifies minimization or maximization of the objective function as an objective associated with compressing the machine learning model. In another example, the user specifies the limit on the accuracy loss as a percentage or proportion of the accuracy of the uncompressed machine learning model. In a third example, the user specifies the compression scheme as quantization, unstructured magnitude-based pruning of weights in a DNN, pruning of neurons in the DNN, pruning of blocks of various dimensions in the DNN, and/or low-rank tensor factorization.

(58) Next, Bayesian optimizer **122** determines **404** a first sparsity ratio associated with the limit on the accuracy loss caused by compressing the machine learning model using the compression scheme **404**. For example, Bayesian optimizer **122** performs a first series of iterations that use an ILS-UCB sampling criterion to sample sparsity ratios for compressing the machine learning model. At the end of the first series of iterations, Bayesian optimizer **122** identifies the first sparsity ratio as the amount of compression (e.g., zero-valued parameters, quantization of parameters, etc.) in the machine learning model that produces an accuracy corresponding to the limit on the accuracy loss.

(59) Bayesian optimizer **122** then selects **406** a second sparsity ratio that optimizes the objective function for the machine learning model within a search space bounded by the first sparsity ratio. For example, Bayesian optimizer performs a second series of iterations that use a GP-UCB (or GP-LCB) sampling criterion to sample sparsity ratios that maximize (or minimize) the objective function, where the sampled sparsity ratios range from 0 to the first sparsity ratio. At the end of the second series of iterations, Bayesian optimizer **122** identifies the second sparsity ratio as the amount of compression in the machine learning model that best optimizes the objective function within exceeding the accuracy loss limit.

(60) During each iteration performed by Bayesian optimizer **122** to select the first and second sparsity ratios, compression engine **124** generates a different compressed version of the machine learning model, where each compressed version has a sparsity ratio selected by Bayesian optimizer **122** for that iteration. Compression engine **124** also determines the accuracy and/or value of the objective function for the sparsity ratio, and Bayesian optimizer **122** updates a GP prior for the accuracy and/or objective function based on the accuracy and/or value of the objective function. The operation of compression engine **124** is described in further detail below with respect to FIG. 5.

(61) Finally, Bayesian optimizer **122** and/or compression engine **124** generate **408** a compressed version of the machine learning model having the second sparsity ratio. For example, Bayesian optimizer **122** samples the second sparsity ratio by optimizing an acquisition function over the GP prior for the objective function. Compression engine **124** then compresses the machine learning model to have the second sparsity ratio and verifies that the compressed machine learning model produces a better value of the objective function than previous compressed versions of the machine learning model.

(62) FIG. 5 is a flow diagram of method steps for compressing a machine learning model, according to one or more aspects of various embodiments. Although the method steps are described in conjunction with the systems of FIGS. 1 and 2, persons skilled in the art will understand that any system configured to perform the method steps in any order falls within the scope of the present disclosure.

(63) As shown, compression engine **124** receives **502** a sparsity ratio sampled during an iteration performed by Bayesian optimizer **122**. For example, the sparsity ratio includes a randomly sampled

sparsity ratio that is used to “seed” the GP prior for a black box function to be optimized and/or a sparsity ratio that is selected by Bayesian optimizer **122** based on sampling criteria and/or an acquisition function associated with the GP prior.

(64) Next, compression engine **124** performs **504** a learning step that trains the machine learning model with a quadratic regularization term. For example, compression engine **124** updates parameters of the machine learning model using a loss function that includes the quadratic regularization term. In turn, the quadratic regularization term imposes a penalty that pulls some parameters to zero based on squares calculated from the values of the parameters.

(65) After the learning step is complete, compression engine **124** performs **506** a compression step that compresses the machine learning model to have the sparsity ratio provided by the Bayesian optimizer. For example, compression engine **124** marks a subset of parameters in the machine learning model to be compressed.

(66) Compression engine **124** additionally repeats operations **504-506** until a stopping criterion is met **508**. For example, the stopping criterion includes convergence in the parameters of the compressed machine learning model. As a result, compression engine **124** iteratively alternates between the learning step and compression step, which allows different sets of marked parameters to be explored until the trained parameters produced by the learning step equal the marked parameters in the compression step. In another example, compression engine **124** repeats operations **504-506** until a fixed number of learning and compression steps have been performed.

(67) Compression engine **124** then provides **510** a measurement of accuracy and/or performance of the compressed machine learning model to Bayesian optimizer **122**. For example, compression engine **124** determines the accuracy of the compressed machine learning model during a first series of iterations performed by the Bayesian optimizer to find a first sparsity ratio that reflects an accuracy loss limit for the compressed machine learning model. Compression engine **124** also, or instead, determines the performance of the compressed machine learning model in optimizing a predefined objective function during a second series of iterations performed by Bayesian optimizer **122** to find a second sparsity ratio that produces the best value of the objective function.

(68) Operations **502-510** may be repeated for remaining iterations **512** performed by Bayesian optimizer **122**. For example, compression engine **124** receives a new sparsity ratio at the beginning of each iteration performed by Bayesian optimizer **122**, creates a compressed machine learning model having the sparsity ratio, and provides the accuracy and/or performance of the compressed machine learning model to Bayesian optimizer **122**. Bayesian optimizer **122** then uses the measured accuracy and/or performance to update a GP prior for the accuracy and/or performance and uses an acquisition function and/or selection criteria for the GP prior to select the next sparsity ratio to sample in a successive iteration. As a result, compression engine **124** repeatedly generates compressed versions of the machine learning model and determines the accuracy and/or performance of each compressed version until a compressed version with the best performance that still maintains an acceptable level of accuracy is found.

(69) In sum, the disclosed embodiments perform sample-efficient Bayesian optimization of sparsity ratios in compressing a machine learning model. A first series of Bayesian optimization iterations is performed to identify a first sparsity ratio that represents a lower limit on the loss in accuracy from compressing the machine learning model. A second series of Bayesian optimization iterations is then performed to identify a second sparsity ratio that produces the best value of a predefined objective function within a search space bounded by the first sparsity ratio. The machine learning model is then compressed to have the second sparsity ratio, which allows the compressed machine learning model to optimize the objective function while maintaining an accuracy that is higher than the lower limit on the accuracy loss.

(70) One technological advantage of the disclosed techniques is the ability to infer a sparsity ratio that meets constraints and/or objectives associated with compressing the machine learning model instead of requiring a user to manually specify a sparsity ratio for compressing the machine

learning model. In turn, the compressed machine learning model incurs significantly less inference time and resource overhead than the uncompressed machine learning model without appreciable loss of accuracy. Another technological advantage includes sample-efficient selection of the sparsity ratio via Bayesian optimization, which significantly reduces the number of iterations (and associated latency and resource consumption) required to identify the optimal sparsity ratio for compressing the machine learning model over conventional techniques that involve manual and/or random searching of sparsity ratios for compressing machine learning models. Consequently, the disclosed techniques provide technological improvements in computer systems, applications, and/or techniques for compressing and/or executing machine learning models.

(71) 1. In some embodiments, a computer-implemented method for compressing a machine learning model comprises determining, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model; selecting, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the machine learning model within a search space bounded by the first sparsity ratio; and generating a compressed version of the machine learning model having the second sparsity ratio.

(72) 2. The computer-implemented method of clause 1, further comprising generating a different compressed version of the machine learning model for each iteration performed by the Bayesian optimizer to select the first and second sparsity ratios.

(73) 3. The computer-implemented method of any of clauses 1-2, wherein generating the different compressed version of the machine learning model comprises performing a learning step that trains the machine learning model with a quadratic regularization term; and performing a compression step that compresses the machine learning model to have a specified sparsity ratio at a given iteration of the Bayesian optimizer.

(74) 4. The computer-implemented method of any of clauses 1-3, wherein the learning step and the compression step are repeated until a stopping criterion is met.

(75) 5. The computer-implemented method of any of clauses 1-4, wherein the Bayesian optimizer performs one or more iterations that randomly sample the first and second sparsity ratios for the machine learning model.

(76) 6. The computer-implemented method of any of clauses 1-5, wherein the Bayesian optimizer performs one or more iterations that select the first sparsity ratio based on an implicit level set upper confidence bound (ILS-UCB) sampling criterion.

(77) 7. The computer-implemented method of any of clauses 1-6, wherein the Bayesian optimizer performs one or more iterations that select the second sparsity ratio based on a confidence bound sampling criterion.

(78) 8. The computer-implemented method of any of clauses 1-7, further comprising receiving the predefined objective function from a user.

(79) 9. The computer-implemented method of any of clauses 1-8, wherein the predefined objective function comprises at least one of an evaluation rate for the machine learning model, a memory footprint, an evaluation time for the machine learning model, and a domain-specific objective.

(80) 10. The computer-implemented method of any of clauses 1-9, wherein the Bayesian optimizer selects the second sparsity ratio based on a Gaussian process prior for the predefined objective function.

(81) 11. The computer-implemented method of any of clauses 1-10, wherein generating the compressed version of the machine learning model comprises at least one of quantizing weights of the machine learning model and pruning a subset of the weights from the machine learning model.

(82) 12. In some embodiments, a non-transitory computer readable medium stores instructions that, when executed by a processor, cause the processor to compress a machine learning model by performing the steps of determining, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model; selecting, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the

machine learning model within a search space bounded by the first sparsity ratio; and generating a compressed version of the machine learning model having the second sparsity ratio.

(83) 13. The non-transitory computer readable medium of clause 12, wherein the steps further comprise generating a different compressed version of the machine learning model for each iteration performed by the Bayesian optimizer to select the first and second sparsity ratios.

(84) 14. The non-transitory computer readable medium of any of clauses 12-13, wherein generating the different compressed version of the machine learning model comprises performing a learning step that trains the machine learning model with a quadratic regularization term; and performing a compression step that compresses the machine learning model to have a specified sparsity ratio at a given iteration of the Bayesian optimizer.

(85) 15. The non-transitory computer readable medium of any of clauses 12-14, wherein the Bayesian optimizer performs one or more iterations that randomly sample the first and second sparsity ratios for the machine learning model.

(86) 16. The non-transitory computer readable medium of any of clauses 12-15, wherein Bayesian optimizer performs a first series of iterations that select the first sparsity ratio based on an implicit level set upper confidence bound (ILS-UCB) sampling criterion and a second series of iterations that select the second sparsity ratio based on a confidence bound sampling criterion.

(87) 17. The non-transitory computer readable medium of any of clauses 12-16, wherein the steps further comprise receiving the predefined objective function and the limit on the accuracy loss from a user.

(88) 18. The non-transitory computer readable medium of any of clauses 12-17, wherein the predefined objective function comprises at least one of an inference throughput, a memory footprint, a latency, and a domain-specific objective.

(89) 19. The non-transitory computer readable medium of any of clauses 12-18, wherein generating the compressed version of the machine learning model comprises at least one of quantizing weights of the machine learning model and pruning a subset of the weights from the machine learning model.

(90) 20. In some embodiments, a system comprises a memory storing instructions; and a processor for executing the instructions to determine, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model; select, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the machine learning model within a search space bounded by the first sparsity ratio; and generate a compressed version of the machine learning model having the second sparsity ratio.

(91) Any and all combinations of any of the claim elements recited in any of the claims and/or any elements described in this application, in any fashion, fall within the contemplated scope of the present invention and protection.

(92) The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments.

(93) Aspects of the present embodiments may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “module,” a “system,” or a “computer.” In addition, any hardware and/or software technique, process, function, component, engine, module, or system described in the present disclosure may be implemented as a circuit or set of circuits. Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(94) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(95) Aspects of the present disclosure are described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine. The instructions, when executed via the processor of the computer or other programmable data processing apparatus, enable the implementation of the functions/acts specified in the flowchart and/or block diagram block or blocks. Such processors may be, without limitation, general purpose processors, special-purpose processors, application-specific processors, or field-programmable gate arrays.

(96) The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(97) While the preceding is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A computer-implemented method for compressing a machine learning model, comprising: determining, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model; selecting, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the machine learning model within a search space bounded by the first sparsity ratio; and generating a compressed version of the machine learning model having the second sparsity ratio.

2. The computer-implemented method of claim 1, further comprising generating a different compressed version of the machine learning model for each iteration performed by the Bayesian optimizer to select the first and second sparsity ratios.
3. The computer-implemented method of claim 2, wherein generating the different compressed version of the machine learning model comprises: performing a learning step that trains the machine learning model with a quadratic regularization term; and performing a compression step that compresses the machine learning model to have a specified sparsity ratio at a given iteration of the Bayesian optimizer.
4. The computer-implemented method of claim 3, wherein the learning step and the compression step are repeated until a stopping criterion is met.
5. The computer-implemented method of claim 2, wherein the Bayesian optimizer performs one or more iterations that randomly sample the first and second sparsity ratios for the machine learning model.
6. The computer-implemented method of claim 2, wherein the Bayesian optimizer performs one or more iterations that select the first sparsity ratio based on an implicit level set upper confidence bound (ILS-UCB) sampling criterion.
7. The computer-implemented method of claim 2, wherein the Bayesian optimizer performs one or more iterations that select the second sparsity ratio based on a confidence bound sampling criterion.
8. The computer-implemented method of claim 1, further comprising receiving the predefined objective function from a user.
9. The computer-implemented method of claim 1, wherein the predefined objective function comprises at least one of an evaluation rate for the machine learning model, a memory footprint, an evaluation time for the machine learning model, and a domain-specific objective.
10. The computer-implemented method of claim 1, wherein the Bayesian optimizer selects the second sparsity ratio based on a Gaussian process prior for the predefined objective function.
11. The computer-implemented method of claim 1, wherein generating the compressed version of the machine learning model comprises at least one of quantizing weights of the machine learning model and pruning a subset of the weights from the machine learning model.
12. A non-transitory computer readable medium storing instructions that, when executed by a processor, cause the processor to compress a machine learning model by performing the steps of: determining, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model; selecting, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the machine learning model within a search space bounded by the first sparsity ratio; and generating a compressed version of the machine learning model having the second sparsity ratio.
13. The non-transitory computer readable medium of claim 12, wherein the steps further comprise generating a different compressed version of the machine learning model for each iteration performed by the Bayesian optimizer to select the first and second sparsity ratios.
14. The non-transitory computer readable medium of claim 13, wherein generating the different compressed version of the machine learning model comprises: performing a learning step that trains the machine learning model with a quadratic regularization term; and performing a compression step that compresses the machine learning model to have a specified sparsity ratio at a given iteration of the Bayesian optimizer.
15. The non-transitory computer readable medium of claim 13, wherein the Bayesian optimizer performs one or more iterations that randomly sample the first and second sparsity ratios for the machine learning model.
16. The non-transitory computer readable medium of claim 13, wherein Bayesian optimizer performs a first series of iterations that select the first sparsity ratio based on an implicit level set upper confidence bound (ILS-UCB) sampling criterion and a second series of iterations that select the second sparsity ratio based on a confidence bound sampling criterion.

17. The non-transitory computer readable medium of claim 12, wherein the steps further comprise receiving the predefined objective function and the limit on the accuracy loss from a user.
18. The non-transitory computer readable medium of claim 12, wherein the predefined objective function comprises at least one of an inference throughput, a memory footprint, a latency, and a domain-specific objective.
19. The non-transitory computer readable medium of claim 12, wherein generating the compressed version of the machine learning model comprises at least one of quantizing weights of the machine learning model and pruning a subset of the weights from the machine learning model.
20. A system, comprising: a memory storing instructions; and a processor for executing the instructions to: determine, by a Bayesian optimizer, a first sparsity ratio associated with a limit on an accuracy loss caused by compressing the machine learning model; select, by the Bayesian optimizer, a second sparsity ratio that optimizes a predefined objective function for the machine learning model within a search space bounded by the first sparsity ratio; and generate a compressed version of the machine learning model having the second sparsity ratio.
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